

Q2

(a) 结果如图所示：

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Call:
lm(formula = Y ~ X1 + X2 + X3 + X4, data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-2.17355 -0.55425 -0.00316  0.61569  2.02727

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.22211    0.71119   17.185 < 2e-16 ***
X1          -0.18698    0.02497   -7.489 9.04e-09 ***
X2           0.29510    0.07349    4.016 0.000298 ***
X3          -1.21196    1.40668   -0.862 0.394786
X4           0.07479    0.01637    4.569 5.86e-05 ***
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9353 on 35 degrees of freedom
Multiple R-squared:  0.7541,    Adjusted R-squared:  0.726
F-statistic: 26.84 on 4 and 35 DF,  p-value: 3.088e-10
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可知拟合模型为：

$$Y = 12.22 - 0.187X_1 + 0.295X_2 - 1.21X_3 + 0.0748X_4.$$

且p值为 3.088×10^{-10} ，远小于 0.05

若 $H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0$ ，则在5%水平下拒绝 H_0

(b) 结果如图（用excel打开，并调整了列宽）

Y	X1	X2	X3	X4	stud_res	stud_del	leverage	dffits	cook
13.5	1	5.02	0.14	12.3	-0.881313929	-0.87843421	0.134697480783791	-0.346581194	0.0241814739426817
12	14	8.19	0.27	10.4079	-0.547027035	-0.54147542	0.147195303866736	-0.224957736	0.0103298008237758
10.5	16	3	0	3.9998	0.11452612011702	0.112899332824193	0.369144511712221	0.0863623876264695	0.00153499025270435
15	4	10.7	0.05	5.7112	0.00184414566906532	0.00181760990386815	0.0838072360601021	0.000549728193308382	6.22178647081519e-08
14	11	8.97	0.07	6	0.908126276106259	0.905794105728871	0.0596414897202634	0.228116658839407	0.0104611035952676
10.5	15	9.45	0.24	10.1385	-2.520284183	-2.745620746	0.149791770651279	-1.152449418	0.223816280221457
14	2	8	0.19	3.13	-0.23986832	-0.236611361	0.100284844719632	-0.078995205	0.00128264376040426
16.5	1	6.62	0.6	24.8172	2.13766363372509	2.25956603764036	0.521945446507885	2.36101564780494	0.997829610842131
17.5	1	6.2	0	21.5	2.42926379281283	2.62589593379008	0.203921760826655	1.32901970647059	0.302334125340312
16.5	8	11.78	0.03	25.1015	0.503208582766019	0.497771708298408	0.0595163180928477	0.125219684416416	0.00320487331494699
17	12	14.62	0.08	29.1264	0.730853920073019	0.725897877276492	0.161808636707235	0.318936912029594	0.0206228970787497
16.5	2	11.55	0.03	20.7549	-0.303250984	-0.299280866	0.0771183693816557	-0.086513697	0.0015369023330664
16	2	9.63	0	8.2	0.775534860739037	0.771029051503939	0.0774203139430094	0.223355299953184	0.0100944737885878
16.5	13	12.99	0.04	35.9665	0.271122005890447	0.267501818290015	0.150003589437346	0.112374822364389	0.00259444275857377
17.225	2	12.01	0.03	26.55	-0.131615597	-0.129753863	0.102320655131538	-0.04380678	0.00039489969103542
17	1	12.01	0	29.9	-0.941069237	-0.939490166	0.141575442817909	-0.381535661	0.0292118421411623
16	1	7.99	0.14	18.9258	0.400112459592981	0.3952601415099	0.0683062842177598	0.107022913367097	0.00234737049548047
14.625	12	10.33	0.12	36.6013	-1.151851331	-1.157426559	0.149202882869637	-0.484695565	0.0465343934620698
14.5	16	10.67	0	34.993	-0.586051309	-0.580473602	0.180198394250926	-0.272146999	0.0150988346461025
14.5	3	9.45	0.03	8.5335	-0.608227275	-0.60266883	0.0592900701070672	-0.151301017	0.00466324264376052
16.5	6	12.65	0.13	23.5932	0.0670676722337183	0.066106867486097	0.0930020864168287	0.0211684851412906	9.22450064906643e-05
16.5	3	12.08	0	13	0.338608468222118	0.334284135919913	0.0924383175343373	0.106685084909086	0.00233561634946645
15	3	10.52	0.05	4.05	-0.008970208	-0.008841144	0.0977591297663444	-0.00291022	1.74369244861739e-06
15	3	9.47	0	4.05	0.270208562533852	0.266598684600172	0.0884697156173451	0.0830557619628635	0.00141726720882688
13	14	11.62	0	4.5959	-0.431894646	-0.426818898	0.127846863187539	-0.163415146	0.00546868571385303
12.5	1	5	0.33	12	-1.776371827	-1.835507218	0.176016821741697	-0.848347943	0.134813560305203
14	15	9.89	0.05	8.1243	1.24756187515221	1.25789708818663	0.0836910290950489	0.380157529495183	0.0284309357761295
13.75	16	11.13	0.06	15.3947	0.174568467954069	0.172131513391033	0.0825225109888382	0.0516236723332873	0.000548199472725382
14	2	7.96	0.22	9.7321	-0.733720368	-0.728789274	0.0795322876507178	-0.214224644	0.00930306514262189
15	16	10.73	0.09	27.6099	0.730635239408156	0.725677314952337	0.10296404391857	0.245856323746104	0.0122548208096471
13.75	2	7.95	0	9	-1.249953962	-1.260421576	0.0866018771602476	-0.388105167	0.0296268324466092
15.625	3	9.1	0	18.4	-0.108416507	-0.106874423	0.0710307882343631	-0.029552651	0.000179748853915078
15.625	3	12.05	0.03	18.4718	-1.043192193	-1.044548682	0.0771826397974737	-0.302086016	0.0182038196656376
13	16	8.43	0.04	9.6	0.688281672133609	0.68301594572351	0.0953118767026667	0.221694494885046	0.00998183847536138
14	16	10.6	0.04	10.635	1.00213417775142	1.00219715470751	0.089198646471183	0.313632069607666	0.0196705426341341
15.25	13	10.55	0.1	13.5512	1.60062662235242	1.63871148376387	0.0580699504026087	0.406882394537172	0.0315895086409703
16.25	1	5.5	0.21	18	1.72152177418598	1.77349666917662	0.132030183103177	0.691695066958755	0.0901620210883455
13	14	8.53	0.03	31.5	-1.693000376	-1.741473039	0.171710923186287	-0.792911561	0.118839777571932
14.5	3	9.04	0.04	4.25	-0.109289486	-0.107735278	0.0753627025834454	-0.03075746	0.000194702628890324
11.5	15	8.2	0	3.0005	-0.640939528	-0.63545716	0.122066804635786	-0.2369487	0.0114235270341974

(c) 6 8 9

(d) 根据 $h_i > 2p/n$, 得到 outlying X Observation 点为3 8

(e) 使用 DFFITS: 6 8 9 26 30

使用 Cook's distance: 不存在这样的点